

URS: Expectations of the product from a user – a broad, qualitative requirement.

<i>User Requirement Specifications</i>			
<u>Category</u>	<u>Specification</u>	<u>Justification</u>	<u>Priority (1-3)</u>
1. Function BN	1.1. The compact device must keep insulin at a safe storage temperature.	Insulin must be stored in an adequate environment to avoid it from going bad. As the design will be used in developing countries with little other access to similar products, the design must ensure that the transported goods remain good. It must do so consistently regardless of exterior temperature.	1 - High
	1.2. The device must be able to transport a large number of insulin doses at once.	As diabetes is an extremely wide and common issue with nearly 70 million people worldwide suffering from it, the design must be able to transport a high number of doses at once to reach remote areas with as few trips as possible.	1 - High
	1.3. The device must be able to transport insulin in a variety of formats.	As insulin can come in the form of vials and epi-pens, the compact device must be able carry in an organised and arranged manner a multitude of insulin forms.	2 - Medium
2. Environmental PK	2.1. Materials used must be recyclable, non-toxic, and environmentally sustainable.	Using materials that are recyclable, non-toxic, and environmentally sustainable ensures that there is a reduced environmental impact during production and waste management. This aligns with global sustainability goals such as net-zero.	1 - High
	2.2. The product should be energy-efficient and use renewable energy sources.	Target regions often have unreliable electricity sources which compromises the storage of insulin. Therefore, designing a product that uses renewable energy supports off-grid use and reduces carbon emissions.	1 - High
	2.3. The product must be suitable for use in hot and humid climate conditions.	In tough environmental conditions the product must be designed to ensure stable operation, preventing insulin from going bad. This makes sure that patients are provided with safe medicine	1 - High
	2.4. The product must have a long operational lifetime.	A product with a long operational lifetime ensures that material waste, manufacturing demand and overall lifecycle costs are reduced. A product that is long-lasting ensures less frequent need for replacements thus reducing the need for more resources. This supports responsible use of materials.	1 - High
3. Ergonomics AT	3.1. The fridge should be lightweight, compact, for carrying over long distances by foot or bicycle.	Lightweight and compact design addresses the frequent need for patients and healthcare workers to traverse long distances in rural or urban LMICs, where walking and public transport are common.	1 - High
	3.2. The fridge must come with a	A comfortable strap reduces stress for the user carrying it for long period of time. An	1 - High

	comfortable and adjustable strap and/or handle.	adjustable strap supports users of different statures, enabling comfortable positioning whether carried over the shoulder, cross-body, or by hand, and accommodates situations where bulky clothing is worn in different climates.	
	3.3. The device shall be simple to open and close securely using a one-handed, intuitive mechanism.	One-handed, intuitive mechanisms ensure usability for individuals who may have physical limitations or who must operate the device in crowded or stressful environments. An intuitive mechanism also avoids confusion for patients from a country with low education index.	1 - High
	3.4. The locking mechanism needs to be completed in a single motion.	A one-motion locking mechanism ensures fast, intuitive, and secure operation for all users. It minimises errors and time spent opening or closing the device, which is critical in emergencies and for those with limited dexterity.	2 - Medium
	3.5. The device shall provide clear, visual temperature indication on its exterior.	Clear, exterior temperature indication helps users quickly verify storage safety, reducing risk of medicine spoilage and ensuring patient safety even for those unfamiliar with technical interfaces.	2 - Medium
4. Product Quality EGW	4.1. The product must be	Must be a high-quality product with all significant aspects making it costly contributing to functionality, durability and reliability rather than 'luxurious' finishes. This will ensure the budget goes to all key areas, rather than unnecessary extras.	
	4.2.Durability	The transporter should be able to last multiple lifespans of the duration sealed, refrigerated insulin can last. Due to it being used as a transporter it must also be assumed to be the only protection available for the insulin vials and jabs. Therefore, it should be able to withstand any impacts to be expected from 'non-careful' transportation. Should also not be prone to corrosion from the elements as it will be used outside on a regular basis and for prolonged periods of time. Any moving parts should be designed to last for a significant period. Any moving parts or systems should be designed with simplicity and durability in mind so that simple repairs can take place with limited resources.	
	4.3.Reliability	Must be designed so that critical cooling elements are as reliable as possible, and backups are in place for reasonable failures.	
5. Safety OS	5.1. A visual display must clearly indicate internal	The users must be able to understand every aspect of the device. This ensures users can quickly assess the safety status. People should be able to understand the expiry dates, internal	1 - High

temperature, battery level, and manufacturing date.

temperature and battery level in critical in low-literacy or low-infrastructure settings. This allows for fast detection and possible prevention of the problem.

5.2. The casing must be impact-resistant and survive a reasonable drop without structural or functional failure.

The system should be able to survive rough terrains or drops from vehicles as the devices primary users are going to be people in underdeveloped and hard to access areas. These locations are also likely to have no infrastructure and proper roads, meaning they should be able to survive drops from trucks, vans and other types of transport.

1 - High

5.3. The enclosure must protect against dust and water ingress (Need a proper IP rating).

The enclosure should be able to keep the insides clean and dry. Dust and other liquids interfere with the hygiene of the medicine. Furthermore, water and other external intrusions can interfere with the internal temperature.

1 - High

5.4. The device must prevent contamination of stored medicines.

Hygienic surfaces and isolation prevent microbial growth and cross-contamination that can damage the insulin.

1 - High

5.5. Battery and charging systems must not overheat, leak, or pose fire risk.

Ensures user safety during transport, especially in hot climates. Overheating and leaks risk damaging insulin vials by changing the internal temperature.

1 - High

5.6. Clear warning system if internal temperature is out of range.

Having a clear warning allows for immediate action to prevent spoilage or unsafe usage of medicine.

1 - High

6. Aesthetics NA

6.1. Product must have a simple functional aesthetic for easy temperature, expiry and status readings.

The aesthetics of the design must reflect the functionality that it serves. Therefore, a very simple approach to the design should be used, to convey the ease of use of its possible displays and storage systems.

2 - Medium

6.2. Product must be easy to handle, compact, and made of durable materials.

Ensures usability and longevity in areas with unreliable transport

2 - Medium

5.4 IP Rating: standart <https://www.iec.ch/ip-ratings?utm>

SRS: Quantifiable, measurable and unambiguous objectives. Must stem from the user requirement specification.

<i>Specific Requirement Specifications</i>			
<u>Category</u>	<u>Specification</u>	<u>Justification</u>	<u>Priority (1-3)</u>
1. Function BN	1.1.a. The compact device must store the insulin between 2-6°C.	According to <i>Diabetes.org</i> , insulin must ideally be kept in a temperature range from 2-6°C to not spoil. And it should no matter what stay below 25°C. Therefore, no matter the environment the insulin is being carried in, the storage chamber must keep the temperature low. As the product is likely to be used in very warm climates, this is an extremely important parameter.	1 - High
	1.1.b. The insulin must never be exposed to direct sunlight.	Insulin loses its potency exposed to direct heat and sunlight. Therefore, the no light should be able to enter the compact fridge in order for the insulin to remain of good quality.	1 - High
	1.2.a. The compact design must be able to carry at least 50 doses on insulin.	Assuming the weight of one vial to be 15g, this would equate to 0.75kg of medicine. As the design must be easily transportable, weight must be considered and balanced with the amount of insulin it can carry.	2 - Medium
	1.3.a. The compact fridge must be able to carry insulin in the form of vials and epi pens.	As the form of insulin varies from vaccines to epi-pen type formats, the compact fridge must be suited to carry all the forms that insulin can come in. The standard vial size is of 25mm of diameter and 45mm of height (NAF, 2025).	1 - High
2. Environmental PK	2.1.a. Manufacturer and production must be ISO 14001 Environmental Management compliant.	Ensuring manufacturers follow internationally recognised standards promotes responsible use of materials and resources ensuring reduced waste and ethical production	1 - High
	2.2.a. Materials must be UV-resistant, corrosion resistant, and durable to maintain a lifespan of 5 years.	Using materials that are resistant to harsh environmental conditions protects the product and insulin from deteriorating. This extends the lifespan of the product by making it more reliable.	1-High

3. Ergonomics AT	2.3.a. The product must have an IP rating of IP 55.	Achieving a high IP rating ensures the product is well-protected against dust and water which is critical to maintaining operation in harsh environments. A high IP rating guarantees the internal electronics are protected from damage. This reduces the chances of the product failing or having to be repaired thus extending its lifespan	1- High
	3.1.a. The fridge's total mass must not exceed 8 kg, overall dimensions must not exceed 25x25x15cm.	All major structural components should be constructed from lightweight, impact-resistant plastics or composite polymers, with density < 0.8 g/cm ³ .	1-High
	3.2.a. The fridge must include an adjustable padded strap, and a handle with soft-touch, sweat-resistant material	The material of the strap must support loads of at least 5 kg without deformation, and the minimum adjustment range of between 30–90 cm.	1-High
	3.3a. The fridge shall utilise a latch, lever, or magnetic closure requiring around 20 N hand force.	According to ADA Standards (Section 309.4): operable parts must be usable with one hand and require no more than 5 lbf (≈22.2 N) to activate, with actuation path clearly marked visually or by tactile cues.	1-High
	3.4.a. Locking and unlocking process must be achievable via a single continuous movement.	Unlatching/releasing of the lid of the insulin fridge need to operate with not more than one motion in a single linear or rotational direction.	2-Medium
	3.5.a. The device exterior must feature a digital or color-coded LED/LCD indicator visible at 2 meters in full daylight	Temperature display accuracy must be ±0.5°C; update interval at least every 60 seconds. The indicator shall remain powered and readable for a minimum of 48 hours on battery, even if cooling function fails.	2-Medium
4. Product Quality EGW	4.1 Quality		

4.2 Durability

Reliability

- On average when sealed and refrigerated insulin can last 18-24 months. Once opened if still refrigerated it can last anywhere between 28-56 days depending on type. Pens can be expected to last for the same duration therefore doesn't present any new issues. Once opened whether it's refrigerated or not has no significant affect on the longevity of the insulin. Based of these time spans the product must be able to last ~72 months (6yrs).
- When carried by hand it can be expected to be at a height of 0.5-1.5m off the ground. If unloaded off the back of a reasonable transporter such as a van it can be expected to be 0.5-0.8m off the ground. If on the back of a motorcycle or animal, it may be 0.9-1.4m off the ground. It's at its highest when stacked up on top of each other with an expected height of 1-1.4m and if on a loading bay this could be up to ~1.8m off the ground if considering an expected stack of 5-7. Taking in these considerations it should protect the contents from any falls up to ~1.8m.
- It's expected to be used in many harsh environments so should be designed to withstand corrosion and weathering including ,rust, UV degradation, humidity, temperature cycles and humidity.
- All moving parts should be expected to last the

72months that the overall product is designed to last. All parts that could be purchased as standardised parts should be done so to ensure durability, unless the team believes it can make in house a better version, to be considered better if more cost effective; stronger and simpler.

- Certain components should be designed with easy replacement in mind to ensure each unit can have a longer lifespan and a smaller environmental and economic impact. This could be achieved by pursuing a modular design in certain aspects of the design.

- The whole design is based off the refrigeration of insulin, therefore if the primary cooling method fails it should either have one back up or be simple to replace/mend with limited resources. Adding more than one back up despite being beneficial for reliability may lead to too great an increase in price or add bulk to the design.

5. Safety OS

5.2.a. Battery pack shall comply with UN 38.3 transport tests; chargers should include fuses.

Having the device comply with UN 38.3 tests prevents fire, explosion, or thermal runaway. This ensures regulatory compliance and user safety. This also prevents battery overheating and interfering with internal temperature.

1 - High

	5.3.a. All internal surfaces shall be made out of easy to sanitize and anti-microbial materials. ISO 13485	Prevents contamination and supports sterile storage requirements. Making it out of anti-microbial materials makes it easy to clean by anyone without training. This also mean people can clean and sanitize it with limited cleaning supplies.	1 - High
	5.4.a. Clear label data (manufacturing + expiry) visible on display.	These labels should be easy to understand by anyone to avoid accidental use of the expired product. This increases accountability.	1 - Medium
6. Aesthetics NA	6.1.a. Surfaces shall be made of non/porous, disinfectant/resistant materials	Fast status checks and inclusivity for low literacy users.	2 - Medium

References

<https://www.diabetes.org.uk/about-diabetes/looking-after-diabetes/treatments/insulin/what-is-insulin>

5.2: <https://www.phmsa.dot.gov/training/hazmat/new-un-requirement-test-summaries>

1.3.a. [https://www.naf-vsm.com/flaconcino-di-iniezione-10ml-10r-trasparente-borosilicato-tipo-vetro-tipo-1.html#:~:text=10%20ml%20injections%20vials%20\(10R,x%2045%20x%201.00%20mm.](https://www.naf-vsm.com/flaconcino-di-iniezione-10ml-10r-trasparente-borosilicato-tipo-vetro-tipo-1.html#:~:text=10%20ml%20injections%20vials%20(10R,x%2045%20x%201.00%20mm.)

2.1 <https://www.bsigroup.com/en-GB/products-and-services/standards/ems-iso-14001-environmental-management-system/>

ISO 14001 - Environmental Management System (EMS) (no date) BSI. Available at: <https://www.bsigroup.com/en-GB/products-and-services/standards/ems-iso-14001-environmental-management-system/> (Accessed: 09 November 2025).

<https://idf.org/europe/media/uploads/sites/2/2023/06/IDF-Insulin-Storage-Doc-PRINT-VERSION-3mm-bleed-201119-FINAL.pdf>

Krämer, L. (no date) *Storage of insulin IDF Europe Awareness Paper*, *idf.org*. Available at: <https://idf.org/europe/media/uploads/sites/2/2023/06/IDF-Insulin-Storage-Doc-PRINT-VERSION-3mm-bleed-201119-FINAL.pdf> (Accessed: 09 November 2025).

2.3 <https://www.iec.ch/ip-ratings>

Ingress Protection (IP) ratings (2025) *iec.ch*. Available at: <https://www.iec.ch/ip-ratings> (Accessed: 06 November 2025).

3.3a <https://www.access-board.gov/ada/guides/chapter-3-operable-parts/>

ADA Standards for Accessible Design, Section 309.4, published by the United States Department of Justice on ADA.gov.

5.3 <https://www.iso.org/iso-13485-medical-devices.html?>